Mathematical Model of an Explainable, Cooperative Smart-Grid Digital Twin

15-Day Forecasting → Network Coordination → Household Demand Response → Priority Pooling → Twin Adapter / Knowledge Graph → LLM Explainability

Final consolidated model. All equations are stated in general symbolic form; deployed parameter values are collected in Tables 1–2 and instantiated only in the worked examples (thread household DHH0016; Kalyanpur, Dhanmondi and Lalbag 132/33 kV substations, Dhaka).

Pipeline. Forecast every substation 15 days ahead → coordinate neighbouring substations (inter-utility energy exchange, battery reserves, capped tariff pass-through) → minimise load shedding and cost while protecting customer QoE and SGO profit → prioritise the high-impact events → ground them in a knowledge graph → explain every decision through an LLM under a hard grounding constraint. The research contribution is **explainability over optimisation**: every number in a final customer-facing explanation traces back to either a real BPDB record or a stated equation.

> ⚠ **Scope disclaimer (read first).** Inter-utility "trading" is a **modelling abstraction**: scheduled inter-tie load transfer between DESCO and DPDC substations over the shared 132 kV ring, priced with an exchange premium. It is **not** a claim about current Bangladeshi regulatory or operational practice. The model answers a counterfactual planning question — *"what would coordinated operation be worth, and how would it be explained?"* — and is stated as such throughout.

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0. Notation, index sets and deployed parameters

0.1 Core sets and symbols

Symbol	Meaning
t	30-minute slot, $t = 1 \dots T$; slot length Δt (h)
$d,\ m$	calendar day $d = 1 \dots D$; calendar month m
k	forecast lead time in days, $k = 1 \dots K$; horizon $K = 15$
$h,\ a$	household $h = 1 \dots H$; appliance a with DR category $c(a) \in \{\text{Critical, Schedulable, Curtailable}\}$
$g \in \mathcal{G}$	socioeconomic tier (lower_middle, middle, upper_middle)
$i, j \in \mathcal{S}$	substations; $\mathcal{L} \subseteq \mathcal{S} \times \mathcal{S}$ exchange tie-lines
$\hat{p}_{d+k\|d},\ p_d$	day-ahead-of-k forecast / actual substation daily peak (MW)

Symbol	Meaning
$\tilde{p}_{i,d+k}$	robust (uncertainty-inflated) planning peak, Eq. 12
s_d	grid-stress indicator (dynamic regime variable, Eq. 20)
$\mu(t)$	TOU multiplier: μ_{pk} peak, μ_{off} off-peak, μ_{std} otherwise
$r_g,\ \alpha_g$	tier retail rate (BDT/kWh) and tier fairness weight
$d_{s,g},\ d_{c,g}$	per-kWh discomfort cost of shifting / curtailing for tier g
$x_{h,t},\ y_{h,t}$	MOLP decisions: shifted fraction of schedulable load; curtailed fraction of curtailable load
$\kappa,\ \rho$	curtailment cap; DR rebate (BDT/kWh) on stress days
ζ_g	tier per-household coincident peak (kW)
k_y	yearly BPDB calibration multiplier (Eq. 6)
$G_i, B_i^{ch}, B_i^{dis}, T_{ij}, LS_i$	network decisions: grid import, battery charge/discharge, $i \to j$ transfer, load shed (MW)
$\mathrm{SoC}_i, \bar{E}_i, \bar{B}_i, R_i$	battery state of charge, energy cap (MWh), power rating (MW), reserve floor (MWh)
π_{ij}^{tr}, m, v	exchange price; premium over supply cost; Value of Lost Load (BDT/kWh)
c_{deg}	battery degradation cost per kWh throughput
$\varphi_i, \delta_{\max}$	customer pass-through share (decision var); max retail surcharge
$\eta \in \{\text{normal, stress, emergency}\}$	Priority-Pool operating regime (Eq. 41)
$\mathcal{P},\ \mathcal{P}_N$	Priority Pool; its top-N truncation
$s(\cdot)$	serialisation operator: structured object → grounded context block
A_θ	Twin Adapter (Eq. 47)

Tier-indexed parameters: wherever a triple is listed (e.g. shift discomfort 2 / 4 / 8) the values apply to (lower_middle / middle / upper_middle) respectively; each carries the tier index g.

0.2 Table 1 — household / data-layer parameters

Parameter	Symbol	Deployed value
Slot length / horizon	$\Delta t;\ T;\ D$	0.5 h; 43,680; 910 (2024-01-01 → 2026-06-28)
TOU multipliers (17–23 / 23–06 / else)	$\mu_{pk} / \mu_{off} / \mu_{std}$	1.35 / 0.80 / 1.00

Parameter	Symbol	Deployed value
Tier retail rate	r_g	6.5 / 8.0 / 10.5 BDT/kWh
Tier fairness weight	α_g	0.80 / 0.60 / 0.35
Shift / curtail discomfort	$d_{s,g};\ d_{c,g}$	2 / 4 / 8; 4 / 7 / 14 BDT/kWh
Curtailment cap; rebate	$\kappa;\ \rho$	0.5; 3 BDT/kWh
SGO supply cost (pk/std/off)	$c_{pk}/c_{std}/c_{off}$	11.5 / 7.5 / 5.5 BDT/kWh
Stress quantile	q_s	0.90 (top 10% of forecasts)
Tier coincident peak	ζ_g	0.35 / 0.95 / 2.60 kW
Tier income bands	—	15–35k / 35–65k / 65–120k BDT
Forecast window; features	$W;\ F$	14 days; 19
L2; learning rate; batch	$\lambda;\ \eta_{lr};\ B$	10^{-4}; 10^{-3}; 16
Early-stopping patience	—	15 epochs (max 150)
Winsorisation quantiles (train)	q_{lo}, q_{hi}	0.005, 0.995
Base GCS weights (normal regime)	$w_1..w_4$	0.40, 0.20, 0.20, 0.20
Appliance-priority values	a_{cu}, a_{sh}	0.7 / 0.4
Pool admission threshold (normal)	γ_{normal}	0.5
LLM payload size	N	500 events
Flexibility base; noise	$F_c;\ \sigma_\varepsilon$	0.05 / 0.70 / 0.45; 0.08
Hot-day threshold	θ_{hot}	33 °C

0.3 Table 2 — network-layer parameters

Parameter	Symbol	Deployed value	Basis
Firm capacity	$\bar{P}_i$	Kalyanpur 140 / Dhanmondi 160 / Lalbag 100 MW	≈ p99–max of real BPDB peaks; Dhanmondi deliberately capacity-short
Tie-line capacity; loss	$\bar{T}_{ij};\ \eta_\ell$	25 MW; 0.03	33 kV inter-tie sizing
Exchange premium	m	2.0 BDT/kWh	> 0 (export profit), $\ll v$
Value of Lost Load	v	60 BDT/kWh (swept 30/60/120)	urban South-Asian order of magnitude
Battery	$\bar{E}_i; \bar{B}_i; R_i$	20 MWh; 10 MW; $0.5\bar{E}_i$	reserve-sized BESS
Round-trip efficiency	η_c, η_d	0.95, 0.95	LFP ≈ 0.90 round-trip

Parameter	Symbol	Deployed value	Basis
Degradation cost	c_{deg}	2.5 BDT/kWh throughput	$C_{batt}/(2N_{cyc}\bar{E})$
Grid cap (normal/nat-peak/outage)	$\bar{G}_i$	$\bar{P}_i$ / $0.85\bar{P}_i$ / 0	scarcity scenarios
Max surcharge; QoE floor	$\delta_{\max}$; $\mathrm{QoE}_{\min}$	0.05; 80	policy params, swept
Horizon; robustness quantile	K; z_β	15 days; $z_{0.9} = 1.28$	receding-horizon planning

> **Dynamic variables** are quantities whose value changes with an index (t, d, m, h, g, i or a regime) and thereby switches the behaviour of an equation; each is flagged where it first drives a decision.

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1. Synthetic data layer — tier-driven load and bills

1.1 Calendar demand factor

$$m_d = m(\text{daytype}_d), \qquad m : \mathcal{D} \to (0, 1] \tag{1}$$

Dynamic variable — m_d switches with $\mathcal{D} = \{\text{Weekday}, \text{Weekend}, \text{Public Holiday}, \text{Durga Puja}, \text{Eid}\}$; deployed mapping (1.00, 0.94, 0.92, 0.95, 0.78). On Eid whole-city demand drops 22% as Dhaka partially empties.

1.2 Diurnal temperature (anchored on real BPDB extremes)

$$T_d(\tau) = T_{d,\min} + (T_{d,\max} - T_{d,\min}) \cdot \tfrac{1}{2}\left[1 - \cos\left(\pi \tfrac{\tau - \tau_1}{\tau_2 - \tau_1}\right)\right], \quad \tau \in [\tau_1, \tau_2] \tag{2}$$

Half-cosine rise on $[\tau_1, \tau_2]$ (deployed 05:30→14:30); a mirrored half-cosine handles the fall. Humidity moves inversely to T around the daily BPDB mean.

1.3 Appliance and household load

$$P_{h,a}(t) = q_{h,a} \cdot p_a \cdot \delta_a \cdot u_a(t) \cdot g_a\big(T_d(t)\big) \tag{3}$$

q = owned quantity, p_a = unit power (W), δ_a = duty cycle, $u_a(t) \in \{0, 1\}$ usage schedule, $g_a(\cdot) \geq 1$ the temperature-response factor for thermal appliances ($g \equiv 1$ otherwise).

$$L_h(t) = \sum_a P_{h,a}(t) = \sum_{c \in \mathcal{C}} L_h^c(t), \qquad \mathcal{C} = \{\text{crit}, \text{sched}, \text{curt}\} \tag{4}$$

Dynamic variable — $g_a(T)$ makes load weather-dynamic: DHH0016's AC runs far longer in June (22 hot days) than January (0) — the root cause the LLM later cites for the bill rise.

Modeling assumption. $g_a(T)$ and m_d are calibrated only against substation-level BPDB peaks (§1.6); no sub-metered appliance ground truth exists for Dhaka. The bottom-up validation bounds aggregate error, not per-appliance error — a stated limitation.

1.4 Tier composition → number of customers

$$N_{res} = \frac{\sigma_{res} \cdot P_{target}}{\sum_{g \in \mathcal{G}} w_g \, \zeta_g}, \qquad N_g = w_g \cdot N_{res} \tag{5}$$

σ_{res} = residential share of substation peak, w_g = tier mix, ζ_g = per-household coincident peak; incomes drawn uniformly within each tier band.

> **Worked example** — Kalyanpur: $\sigma_{res} = 0.66$, $P_{target} = 96.6$ MW → 63.76 MW residential; $\sum w_g\zeta_g = 0.35{\cdot}0.35 + 0.50{\cdot}0.95 + 0.15{\cdot}2.60 = 0.9875$ kW → $N_{res} \approx$ 64,600, lower_middle ≈ 22,600.

1.5 Yearly BPDB calibration (dynamic count multiplier)

$$k_y = \frac{P_y^{BPDB}}{P_y^{achieved}}, \qquad N_{g,y}^{cal} = k_y \cdot N_g \tag{6}$$

k_y scales the **customer count** per year so the synthetic aggregate reproduces that year's real BPDB mean daily peak — calibration by counts, never by rescaling load profiles.

> **Worked example** — Kalyanpur 2024: $k = 96.6/77.2 = 1.251$ lifts lower_middle 22,619→28,296; k changes yearly (1.251→1.212→1.289).

1.6 Bottom-up validation

$$\text{MAPE} = \frac{100}{D} \sum_{d=1}^{D} \frac{|p_d^{syn} - p_d|}{p_d} \tag{7}$$

> **Worked example** — Kalyanpur: MAPE = 10.04%, Pearson $r = 0.803$ over 811 matched days — no post-hoc scaling.

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2. Prediction layer — 15-day rolling-horizon substation peak

Architecture label: the "Deep Forecasting Model". Six candidate architectures (BiLSTM, BiGRU, BiLSTM+BiGRU, BiGRU+BiLSTM, CNN+BiLSTM+BiGRU, CNN+BiGRU+BiLSTM) are compared under a leak-free protocol; the deployed model is the validation winner, ***BiGRU****. Read any legacy "LSTM Forecasting" label in architecture diagrams as this block.*

2.1 Cleaning and leak-free preprocessing

$$p_d \leftarrow \text{NaN if } \mathcal{A}(p_d) \vee p_d > P_{\max}; \qquad p_d \leftarrow \text{clip}\big(p_d; Q_{q_{lo}}^{train}, Q_{q_{hi}}^{train}\big) \tag{8}$$

Clock-time artefact predicate $\mathcal{A}$; plausibility bound $P_{\max}$. All imputation medians and clipping quantiles come from **train years only**; interpolation is forward-only (causal).

2.2 Features, scaling, windows

$$x_d \in \mathbb{R}^F : \text{ load } p_d, \text{ weather, calendar encodings, lags } p_{d-1}, p_{d-7}, \text{ rolling stats} \tag{9}$$

$$c_{\sin}(v, P) = \sin \tfrac{2\pi v}{P}, \quad c_{\cos}(v, P) = \cos \tfrac{2\pi v}{P}; \qquad \tilde{x} = \frac{x - x^{train}_{\min}}{x^{train}_{\max} - x^{train}_{\min}} \tag{10}$$

$$X_d = [\tilde{x}_{d-W}, \ldots, \tilde{x}_{d-1}] \in \mathbb{R}^{W \times F} \tag{11}$$

> **Worked example** — strict chronological split: train 2019–2023 → 1,521 sequences; validation 2024 → 366; test 2025–26 → 544 ($W = 14, F = 19$). Validation/test windows seed only from past data.

2.3 Recurrent cells (winner: BiGRU)

$$z_t = \sigma(W_z x_t + U_z h_{t-1}), \qquad r_t = \sigma(W_r x_t + U_r h_{t-1}) \tag{12a}$$

$$\tilde{h}_t = \tanh\big(W_h x_t + U_h(r_t \odot h_{t-1})\big), \qquad h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \tag{12b}$$

$$h^{bi}_t = [\overrightarrow{h}_t; \overleftarrow{h}_t] \tag{12c}$$

LSTM candidates add a cell state: $c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, h_t = o_t \odot \tanh(c_t)$.

2.4 Direct multi-horizon head (15-day)

$$\hat{\mathbf{p}}_d = (\hat{p}_{d+1|d}, \ldots, \hat{p}_{d+K|d}) = W_2 \, \mathrm{ReLU}(W_1 h^{bi}_{last} + b_1) + b_2 \in \mathbb{R}^K \tag{13}$$

$$J(\theta) = \frac{1}{NK} \sum_d \sum_{k=1}^{K} \omega_k \big(\hat{p}_{d+k|d} - p_{d+k}\big)^2 + \lambda \|\theta\|^2 \tag{14}$$

Direct multi-horizon is preferred over recursive rollout (which compounds its own errors); both are compared under the leak-free protocol. Future covariates at lead k use only information available at day d (calendar known; weather uses climatological normals beyond the reliable range).

Why $K = 15$ (not 7 or 30). Fifteen days is the shortest horizon that spans a full weekly demand cycle *plus* a second week of pre-positioning lead time for battery reserve and scheduled inter-tie transfers, while remaining inside the range where BiGRU retains usable skill on the BPDB series (§2.5). Seven days gives no pre-positioning margin for a mid-horizon deficit; thirty days pushes well past the point where σ_k swamps the signal, making distant decisions pure noise. $K = 15$ is the planning/skill compromise, and the per-horizon skill curve (§2.5) is reported to justify it empirically rather than by assertion.

2.5 Horizon-dependent uncertainty and robust peaks

$$\sigma_k = \mathrm{sd}\big(\{p_{d+k} - \hat{p}_{d+k|d}\}_{d \in val}\big), \qquad \sigma_1 \le \sigma_2 \le \cdots \le \sigma_K \text{ (empirically)} \tag{15}$$

$$\tilde{p}_{i,d+k} = \hat{p}_{i,d+k|d} + z_\beta \, \sigma_{i,k} \tag{16}$$

Reporting requirement: $R^2(k)$, RMSE(k), σ_k reported for $k = 1 \ldots 15$ as a headline figure. Strong skill is expected at $k \leq 3$, substantial decay by $k \geq 10$; the widening margin $z_\beta \sigma_k$ is precisely what motivates holding battery reserve rather than acting on distant point forecasts.

2.6 Training objective, metrics, global selection

$$R^2 = 1 - \frac{\sum_d (y_d - \hat{y}_d)^2}{\sum_d (y_d - \bar{y})^2}, \quad \text{RMSE} = \sqrt{\tfrac{1}{N}\sum_d (y_d - \hat{y}_d)^2}, \quad \text{MAE} = \tfrac{1}{N}\sum_d |y_d - \hat{y}_d| \tag{17}$$

$$M^* = \arg\max_{M \in \mathcal{M}} \frac{1}{S}\sum_{s=1}^{S} R^2_{val,s}(M) \quad \text{(ties: lowest mean validation rank)} \tag{18}$$

Model	mean Val R²	mean Test R²	wins (val)	decision
BiGRU	**0.6882**	0.7594	9/18	**SELECTED (M^*)**
BiLSTM+BiGRU	0.6815	0.7534	—	—
BiGRU+BiLSTM	0.6784	0.7520	—	old-protocol winner
CNN+BiLSTM+BiGRU	0.6696	0.7405	—	—
CNN+BiGRU+BiLSTM	0.6693	0.7283	—	—
BiLSTM	0.6656	0.7379	—	—

> **Worked example** — deployed BiGRU ($S = 18$): Kalyanpur val R² 0.828 / test 0.853 (MAE 5.23 MW); Dhanmondi val 0.589 / test 0.786. One-step-ahead 2024-06-30 (Dhanmondi): $\hat{p} = 158.91$ MW — the number that triggers everything downstream.

Digital-twin state note. The forecast $\hat{p}_d$ **is** the twin's predictive state; the network LP and household MOLP run on this twin state, and the knowledge graph (§8) is the twin's *explanatory* state, updated after each decision so every action is reconstructible. This is an explainability twin, not a closed-loop control twin.

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3. Network coordination layer — receding-horizon LP

At each day d the SGO coalition solves an LP over the window $k = 1 \ldots K$ (one representative evening-peak block per day keeps the LP small; §11 gives sizes). **Only the $k = 1$ decisions are executed**; the window rolls forward and re-solves with updated forecasts — model-predictive control.

$$s_{i,d+k|d} = 1\left[\tilde{p}_{i,d+k} \ge \tau_i\right], \qquad \tau_i = Q_{q_s}\left(\{\hat{p}_{i,d}\}\right), \qquad \mathrm{Def}_{i,d+k} = \max\left(0, \tilde{p}_{i,d+k} - \bar{P}_i\right) \tag{19}$$

Dynamic variable — the stress indicator is now a *vector over the horizon*: the SGO sees "day 5 has a predicted 12 MW deficit", enabling pre-positioning. **Only the day-ahead value $s_{i,d+1|d}$ (written s_d) propagates to the household MOLP, GCS and pooling layers**; deeper leads inform only battery/transfer scheduling.

3.2 Energy balance (substation i, executed block t)

$$G_i(t) + \eta_d B_i^{dis}(t) - B_i^{ch}(t) + \sum_j (1-\eta_\ell) T_{ji}(t) - \sum_j T_{ij}(t) + LS_i(t) + \Delta_i^{DR}(t) = \tilde{D}_i(t) \tag{20}$$

Demand met by: own grid import, battery discharge, neighbour imports (net of loss), household DR, and — last — load shedding.

3.3 Constraints

$$0 \le G_i(t) \le \bar{G}_i(t), \qquad 0 \le T_{ij}(t) \le \bar{T}_{ij}, \qquad LS_i(t) \ge 0 \tag{21a}$$

$$\mathrm{SoC}_i(t{+}1) = \mathrm{SoC}_i(t) + \eta_c B_i^{ch}(t)\Delta t - B_i^{dis}(t)\Delta t, \quad 0 \le B_i^{ch}, B_i^{dis} \le \bar{B}_i, \quad 0 \le \mathrm{SoC}_i \le \bar{E}_i \tag{21b}$$

$$\mathrm{SoC}_i(t) \ge R_i - u_i(t), \qquad u_i(t) \ge 0 \text{ penalised at } v^{res} \lesssim v \tag{21c}$$

$$0 \le \Delta_i^{DR}(t) \le \Delta_i^{DR,\max}(t) \tag{21d}$$

Battery as operational reserve (21c). The reserve floor R_i may be violated (slack u_i) only at a penalty just below VoLL — so the optimiser dips into reserve *only when the alternative is shedding*. With the degradation cost below, the battery never behaves as a daily-arbitrage asset.

3.4 Battery degradation and exchange price

$$c_{deg} = \frac{C_{batt}}{2N_{cyc}\bar{E}}, \qquad \mathrm{cost}_i^{deg} = c_{deg}\sum_t \left(B_i^{ch}(t) + B_i^{dis}(t)\right)\Delta t \tag{22}$$

$$\pi_{ij}^{tr}(t) = c_i(t) + m, \qquad c_i(t) \in \{c_{off}, c_{std}, c_{pk}\} \tag{23}$$

The exporter is paid marginal supply cost **plus premium m** — export is strictly profitable, import strictly cheaper than shedding ($\pi^{tr} \ll v$), so cooperation is individually rational for both utilities. Export additionally requires headroom $\sum_j T_{ij}(t) \le \bar{P}_i - \tilde{p}_i(t) + \eta_d B_i^{dis}(t)$.

3.5 Objective — split into network cost and penalty cost

For clarity the objective is decomposed (identical LP, easier to defend):

$$C^{pen} = \sum_{i,t} \Big[\underbrace{d_i^{DR} \Delta_i^{DR}}_{\text{DR discomfort}} + \underbrace{v\, LS_i}_{\text{shedding}} + \underbrace{v^{res} u_i}_{\text{reserve dip}} \Big] \Delta t \tag{24b}$$

$$\min_{G,B,T,LS,\Delta^{DR}} \quad C^{net} + C^{pen} \tag{24c}$$

Because $v \gg \pi^{tr} > c >$ (degradation-adjusted battery cost) in the deployed values, the LP realises the intended merit order **through prices alone** — no if-else rules: own grid < neighbour import < battery reserve < shedding. Shedding is the numeric last resort, not an enumerated rule. All variables continuous, all constraints linear → a genuine LP. *(This is the network-coupled problem; the per-household bang-bang result of §4 is preserved at the household level.)*

3.6 Congestion-time allocation priority

When aggregate demand for a scarce tie-line exceeds capacity ($\sum_j \mathrm{Def}_j > \bar{T}_{ij}$), the shed term is disaggregated by a **category-differentiated VoLL** so critical infrastructure is protected first:

$$v_c = v \cdot \chi_c, \qquad \chi_{\text{crit-infra}} > \chi_{\text{residential}} > \chi_{\text{deferrable}} \tag{25}$$

with deployed $\chi = (5.0,\ 1.0,\ 0.4)$ for (hospital/critical, residential, deferrable-commercial). The LP then minimises $\sum_c v_c LS_{i,c}$, automatically routing scarce imported MW to the highest-v_c load — hospitals before deferrable commercial — without an explicit priority rule. *(Category-level extension; full feeder-level rationing is future work.)*

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4. Household MOLP layer — sequential hierarchical follower

The household problem is the **follower** of a sequential hierarchical optimisation (*not* classical bi-level: the follower is not solved inside the leader's constraints — the leader dispatches an aggregate DR quantity computed by the follower, and the follower disaggregates it). The link is one aggregate quantity per substation.

4.1 Prices, decisions, constraints

$$\pi_h(t) = \left(r_{g(h)} + \Delta r_{i,m}\right) \cdot \mu(t), \qquad 0 \le x_{h,t} \le 1, \quad 0 \le y_{h,t} \le \kappa, \quad L_h^{crit} \text{ must-run} \tag{26}$$

where $\Delta r_{i,m}$ is the monthly surcharge from §5 (zero absent any import). The problem is per-household decoupled — no household-coupling constraints — which is what makes the exact bang-bang solution below possible.

4.2 Objectives (household) and bang-bang optimum

$$f_1(\text{cost}) = \sum_{h,t} L_h(t) \pi_h(t) \Delta t - \rho \sum_{h,t} E_{h,t}^{cu} s_{d(t)} \ \rightarrow \ \min \tag{27}$$

$$f_2(\text{discomfort}) = \sum_{h,t} \left(d_{s,g} E^{sh}_{h,t} + d_{c,g} E^{cu}_{h,t} \right) \ \rightarrow \ \min \tag{28}$$

$$U_h = \alpha_g \cdot \Delta B_h - (1 - \alpha_g) \cdot D_h \tag{29}$$

U_h is **linear** in x, y under box constraints ⇒ optimum at a bound (bang-bang); the thresholds are the exact solution and $y^* \in \{0, \kappa\}$ (interior never optimal):

$$x^*_{h,t} = 1 \iff \alpha_g r_g (\mu_{pk} - \mu_{off}) > (1 - \alpha_g) d_{s,g} \tag{30}$$

$$y^*_{h,t} = \kappa \iff \alpha_g \left(r_g \mu_{pk} + \rho s_d \right) > (1 - \alpha_g) d_{c,g}, \qquad \alpha^*_g = \frac{d_{c,g}}{r_g \mu_{pk} + \rho + d_{c,g}} \tag{31}$$

> **Worked example (shift)** — lower_middle $0.8 \cdot 6.5 \cdot 0.55 = 2.86 > 0.40$ → shift; middle $2.64 > 1.60$ → shift; upper_middle $2.02 < 5.20$ → do nothing (saving 0.00%, QoE 100). **Worked example (curtail, stress day)** — upper_middle $0.35(10.5 \cdot 1.35 + 3) = 6.01 < 0.65 \cdot 14 = 9.10$ → no; flip point $\alpha^* = 14/31.175 = 0.449$.

(The SGO/profit objective f_3 is not scalarised here; it is the system-level objective realised by the network LP of §3 and reported ex-post — see §5, Eq. 37.)

4.3 Aggregation link to the network layer

$$\Delta_i^{DR,\max}(t) = \sum_{h \in i} \left(L_h^{sched}(t) 1[x^*_{h,t}{=}1] + \kappa L_h^{curt}(t) 1[y^*_{h,t}{=}\kappa] \right) \tag{32}$$

$$d_i^{DR} = \frac{\sum_{h \in i} \left(d_{s,g(h)} E_h^{sh} + d_{c,g(h)} E_h^{cu} \right)}{\sum_{h \in i} \left(E_h^{sh} + E_h^{cu} \right)} \quad \text{(mean DR discomfort price, BDT/kWh)} \tag{33}$$

The network LP treats DR as one dispatchable resource with capacity (32) and price (33); the bang-bang rules (30–31, with the surcharged tariff of Eq. 26) then disaggregate the dispatched Δ_i^{DR}. Because the surcharge *raises* r_g in the thresholds, pass-through mechanically increases DR willingness — a self-stabilising feedback the LLM can cite.

4.4 Energy accounting, bills, comfort

$$E^{sh}_{h,d} = \sum_{t \in pk(d)} L_h^{sched}(t) x^*_{h,t} \Delta t; \qquad B^{opt}_{h,m} = \sum_{t \in m} L_h^{opt}(t) \pi_h(t) \Delta t - \rho \sum_{t \in m} E^{cu}_{h,t} s_{d(t)} \tag{34}$$

$$S_{h,m} = B^{base}_{h,m} - B^{opt}_{h,m}, \quad \text{QoE}_{h,m} = 100 \left(1 - \frac{D_{h,m}}{\max(B^{base}_{h,m}, 1)} \right) \tag{35}$$

> **Worked example** — DHH0016 (middle), June 2026: $B^{base} = 4{,}647.5$, $B^{opt} = 3{,}924.4$ → $S = 723.0$ BDT. May 2026: $D = 4 \cdot 79.95 + 7 \cdot 27.41 = 511.6$ → QoE $= 100(1 - 511.6/4{,}042.4) = 87.3$. Same equations gave $S = 195.5$ BDT in Jan 2025 (0 hot days) — bill-weather coupling of Eq. 3 propagating into savings.

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5. Premium allocation — who pays to avoid load shedding

Imported energy costs the deficit SGO an **exchange premium** above own-grid cost:

$$\Delta C^{tr}_{i,m} = \sum_{t\in m}\sum_{j} \big(\pi^{tr}_{ji} - c_i(t)\big) T_{ji}(t)\Delta t \tag{36}$$

The split is a **decision variable** $\varphi_i \in [0,1]$: customers pay $\varphi_i \Delta C^{tr}$ as a surcharge, the SGO absorbs the rest:

$$\Delta r_{i,m} = \frac{\varphi_i \Delta C^{tr}_{i,m}}{E^{served}_{i,m}}\ (\mathrm{BDT/kWh}), \qquad \text{s.t. } \Delta r_{i,m} \le \delta_{\max} r_{g(h)},\ \ \mathrm{QoE}_{h,m} \ge \mathrm{QoE}_{\min}\ \forall h \tag{37}$$

$$\varphi_i^* = \arg\min_{\varphi\in[0,\varphi_i^{cap}]} \bar{\alpha}_i \varphi \Delta C^{tr}_i - (1-\bar{\alpha}_i)\big(f_3^{(i)}(\varphi) - f_3^{(i)}(0)\big), \quad \varphi_i^{cap} = \min\left(1, \frac{\delta_{\max}\bar{r}_i E^{served}_i}{\Delta C^{tr}_i}\right) \tag{38}$$

Linear in φ ⇒ **bang-bang at a bound** (same philosophy as §4): $\varphi_i^* = 0$ (SGO absorbs small premiums) or φ_i^{cap} (capped pass-through when a large premium threatens $f_3 \ge 0$). The SGO per-utility objective, reported ex-post along the (α, φ, m) sweep:

$$f_3^{(i)} = \sum_t \Big[\mathrm{Rev}_i - c_i G_i - \sum_j \pi^{tr}_{ji} T_{ji} + \sum_j \pi^{tr}_{ij} T_{ij} - c_{deg}(B_i^{ch} + B_i^{dis}) - \rho E_i^{cu} s - w S_i\Big]\Delta t \tag{39}$$

Derived merit order: cheap own-grid → neighbour import (SGO-absorbed premium) → neighbour import (capped customer surcharge, never above $\delta_{\max}$, never below $\mathrm{QoE}_{\min}$) → battery reserve → controlled shedding at VoLL. Every rung is a numeric consequence of Eqs. 24 and 38. Export revenue ($+\sum_j \pi^{tr}_{ij} T_{ij}$) is the term that makes the surplus utility profit from cooperation, stabilising the coalition.

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6. Adaptive Priority Pooling — Grid Criticality Score

6.1 Component features

$$\mathrm{PDR}_{h,d} = \mathrm{clip}\left(\frac{p^{base}_{h,d} - p^{opt}_{h,d}}{\max(p^{base}_{h,d}, 1)}; 0, 1\right) \tag{40a}$$

p^{base} is the **simulated counterfactual baseline** (load without DR), not a household forecast — deep forecasting exists only at substation level. PDR is the throttle magnitude.

$$\mathrm{PMN}_d = \frac{1 + s_d}{2} \in \{\tfrac{1}{2}, 1\}, \qquad \mathrm{AA}_{h,d} \in \{a_{cu}, a_{sh}\} \tag{40b}$$

$$\mathrm{NI}_e = \mathrm{norm}\Big(\underbrace{\omega_a\, \mathrm{AvoidedShed}_e}_{\mathrm{MWh}} + \omega_t\, \mathrm{TransferMag}_e + \omega_d\, \mathrm{DownstreamCust}_e + \omega_c\, \mathrm{CritInfra}_e\Big) \in [0,1] \tag{40c}$$

NI (Network Impact) is graph-derived: avoided-shedding MWh, transfer magnitude, downstream customers affected, and a critical-infrastructure flag — all read directly from the knowledge graph (§8). This term is what makes a TransferEvent outrank a washing-machine shift, *independent of the weights*.

6.2 Regime-indexed adaptive weights

The score is linear; the weights are a function of a **single discrete operating regime** η that the pipeline already computes (from s_d and network state) — not a learned continuous function:

$$\mathrm{GCS}_e = w_1(\eta)\mathrm{PDR} + w_2(\eta)\mathrm{PMN} + w_3(\eta)\mathrm{AA} + w_4(\eta)\mathrm{NI}, \qquad \sum_k w_k(\eta) = 1 \tag{41}$$

$$\eta = \begin{cases} \text{normal} & s_d = 0 \ \wedge \ \text{no deficit} \\ \text{stress} & s_d = 1 \ \wedge \ \text{no active transfer} \\ \text{emergency} & \bar{G}_i < \tilde{D}_i \ \vee \ \sum_j T_{ij} > 0 \ \vee \ u_i > 0 \end{cases} \tag{42}$$

regime η	**trigger (already computed)**	w(PDR, PMN, AA, NI)	$\gamma(\eta)$
normal	$s_d = 0$, no deficit	0.40, 0.20, 0.20, 0.20	0.50
stress	$s_d = 1$	0.30, 0.25, 0.15, 0.30	0.50
emergency	grid cap < demand ∨ transfer > 0 ∨ reserve dip	0.20, 0.20, 0.10, 0.50	0.55

Design choice — why regime-indexed, not learned. (i) *Explainability*: a 3-row lookup keyed on observable state is auditable and traces to a stated table; a learned softmax over a latent vector is the black box the thesis differentiates itself against. (ii) *No labels*: there is no ground-truth "correct" priority weight to train against. (iii) *No double-counting*: NI already rises during emergencies via its feature values, so the weight shift is a deliberate, bounded amplification rather than a second, hidden state-dependence. (iv) *Footprint*: reorders a 152k-event list with a config block, not a training pipeline. The weights adapt to regime; the *adaptivity to magnitude* is carried by the features — mirroring how the MOLP is "dynamic" through moving prices under fixed α_g.

6.3 Pool admission (regime-consistent threshold)

$$\mathcal{P} = \{e : \ s_d = 1 \ \wedge \ \mathrm{GCS}_e > \gamma(\eta)\} \tag{43}$$

γ is **regime-indexed** so that admission remains comparable across regimes despite the shifting weights (an emergency-regime GCS is scored on a different mix, so its threshold is set accordingly — this closes the cross-regime comparability gap that fixed γ would open). Network-emergency events (transfers, reserve dips, shed-averted) enter the pool directly via the $s_d = 1$ gate and their high NI.

$$\text{context reduction} = 1 - \frac{|\mathcal{P}|}{|\mathcal{E}|} \tag{44}$$

> **Worked example (household, normal/stress)** — DHH0016, 2024-06-30: PMN=1, AA= a_{cu}=0.7, PDR=0.779, NI≈0.15 → η = stress, GCS $= 0.30{\cdot}0.779 + 0.25{\cdot}1 + 0.15{\cdot}0.7 + 0.30{\cdot}0.15 = 0.63 > 0.50$ → enters pool. **Worked example (network, emergency)** — Kalyanpur→Dhanmondi 12.8 MW transfer averting 2.1 h shed: NI≈0.92, PMN=1, η = emergency, GCS $= 0.20{\cdot}0.4 + 0.20{\cdot}1 + 0.10{\cdot}0.7 + 0.50{\cdot}0.92 = 0.81 > 0.55$ → enters pool and outranks the household event, exactly as intended.

6.4 Pool → Prompt transformation

$$\mathcal{P}_N = \mathrm{TopK}_N(\mathcal{P}; \mathrm{GCS}), \qquad N = 500 \tag{45}$$

Each $e \in \mathcal{P}_N$ is serialised into a grounded block $b(e) = s\big(e, \mathrm{grid}(e), \mathrm{molp}(e), \mathrm{network}(e), \mathrm{household}(e)\big)$ containing grid context (substation, date, TOU, stress flag, $\hat{p}$), MOLP decision, network context (transfer, battery, surcharge), and household context. The prompt for query q is

$$\mathrm{Prompt}(q) = f\big(q,\ s(G_q),\ \{b(e) : e \in \mathcal{P}_N \cap G_q\}\big) \tag{46}$$

Retrieval order is Pool → KG → serialise → adapter → LLM: the graph supplies context *before* generation (retrieval-augmented generation). The 89.99%+ pool reduction plus fixed N keep the LLM context bounded as horizon or household count grows.

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7. Twin Adapter / LLM reasoning layer

7.1 Twin Adapter

$$Z_q = A_\theta\big(s(\mathcal{P}_N),\ s(G_q)\big) \tag{47}$$

A_θ has two components: (i) a **deterministic serialisation / prompt-assembly stage** (the payload builder producing grounded JSON blocks with an explicit grounding instruction) — fully implemented; and (ii) an optional **parameter-efficient adaptation** W_A (adapter/LoRA weights fine-tuned on domain state→explanation pairs) so the frozen base model reads twin-state serialisations natively — the framework's fine-tuning pathway.

7.2 Retrieval and grounded generation

$$G_q = \mathcal{N}_k(\mathrm{resolve}(q)) \subseteq G; \qquad Y = \mathrm{LLM}(q, Z_q) \tag{48}$$

$$:e=(u,v)\in E_q\big\} \tag{49}$$

$$\mathrm{nums}(Y) \subseteq \mathrm{vals}(G_q) \cup \mathrm{derived}\big(\mathrm{vals}(G_q)\big) \tag{50}$$

Grounding constraint (50): every number in the answer must be a graph value or arithmetic on graph values. The stored explanation string is a debugging/template fallback — the LLM composes from structured fields.

Causality. The LLM is a language model, not a causal model. Causal structure lives in the **knowledge graph**: directed evidence paths (ForecastDay → stress → TransferEvent/DrEvent → MonthlyBill) encode the decision chain. The LLM **traverses and**

verbalises these stored causal paths under Eq. 50 — it reconstructs "why", it does not infer causation from correlations.

> **Worked example** — "why did my bill carry a surcharge in June?" → G_q returns Dhanmondi day-5 robust forecast 172.4 MW > firm capacity 160; import 12.4 MW from Kalyanpur at 13.5 BDT/kWh (premium 2.0); $\varphi^* = \varphi^{cap}$; surcharge capped at 5% of tier rate; avoided 2.1 h shed at VoLL 44× the surcharge — full causal chain in one retrieval.

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8. Knowledge-graph layer

$$G = (V, E, \ell, \rho), \qquad V = V_{User} \cup V_{Meter} \cup V_{Appliance} \cup \cdots \cup V_{Battery} \cup V_{TransferEvent} \tag{51}$$

Deployed household instance: $|V| = 13{,}301$ nodes / 12 types; $|E| \approx 20{,}750$ / 13 types. **Network extension** adds node types `Battery` (SoC, reserve floor, throughput), `TransferEvent` (MW, price, premium, from/to, avoided-shed MWh), `DeficitForecast` (lead $k, \tilde{p}, \bar{P}$, margin), `SurchargeRecord` ($\varphi, \Delta r$, cap), and edge types `CONNECTED_TO`, `IMPORTED_FROM`/`EXPORTED_TO`, `AVOIDED_SHEDDING`, `RESERVED`/`RESERVE_DIPPED`, `PASSED_THROUGH`.

8.1 Flexibility index (edge property on BELONGS_TO)

$$\varphi_a = F_{c(a)} \cdot \varepsilon, \qquad \varepsilon \sim \mathcal{N}(1, \sigma_\varepsilon^2) \tag{52}$$

> DHH0016's AC: $0.70\varepsilon = 0.753$; refrigerator $0.05\varepsilon = 0.046$ — a 16× ratio the LLM cites when explaining why the AC was chosen and the fridge protected.

8.2 Weather aggregates (MonthlyBill nodes) and forecast evidence (DrEvent/ForecastDay)

$$\text{hot\_days}_m = \sum_{d \in m} 1[T_{d,\max} \geq \theta_{hot}]; \qquad e_d = \hat{p}_d - p_d, \quad \text{margin}_d = \tau - \hat{p}_d \tag{53}$$

The rebate and surcharge are materialised as numeric properties (rebate $= \rho E^{cu}$), so the LLM never parses prose for numbers.

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9. What-if simulation and offline feedback

$$S' = \text{Sim}(S, \Delta), \qquad \Delta \in \{\text{generator outage, grid-cap drop, demand spike, price change, tie-line loss}\} \tag{54}$$

$$\text{Forecast}(S') \rightarrow \text{Network LP}(S') \rightarrow \text{MOLP}(S') \rightarrow \text{GCS}(S') \tag{55}$$

Scenarios must respect the physical constraints of §1/§3 (must-run critical load, box constraints, TOU, tie capacities) — realism is a constraint on Δ, not a free simulation.

$$D_{new} = D \cup D_{syn}(\Delta), \qquad \theta_{t+1} = \text{Train}(D_{new}) \tag{56}$$

Offline batch augmentation, not online retraining: rare-event scenarios are appended to the training corpus and the forecaster periodically re-trained under the same leak-free protocol; no gradient step at inference, so zero serving latency. What-if analysis thereby identifies vulnerable nodes (raising their NI/GCS relevance), enriches forecaster training data, and stress-tests network + MOLP decisions.

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10. Explainability metrics

Explainability is the contribution, so it is measurable rather than implicit. For answer Y to query q with evidence G_q:

$$\mathrm{GF}(Y) = \frac{|\mathrm{nums}(Y) \cap (\mathrm{vals}(G_q) \cup \mathrm{derived})|}{|\mathrm{nums}(Y)|} \in [0, 1] \quad \text{(Grounded-Explanation Ratio; hallucination rate} \tag{57}$$

$$\mathrm{CC}(Y) = \frac{|\{e \in C_q :\ e \text{ cited in } Y\}|}{|C_q|} \quad \text{(Citation / Causal Coverage over the decision path } C_q\text{)} \tag{58}$$

$$\mathrm{ExplainabilityScore}(Y) = \omega_1 \mathrm{GF} + \omega_2 \mathrm{CC} + \omega_3 \mathrm{Faithfulness}, \qquad \textstyle\sum_i \omega_i = 1 \tag{59}$$

These are **evaluated, not optimised**: GF = 1 by construction (hard grounding, Eq. 50); CC and Faithfulness are reported empirically over a GraphRAG query set that now includes **network-coordination queries** (who imported, who paid, why nobody was shed, why the battery held) alongside household queries.

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11. Computational complexity and honest caveats

Per-component cost. Forecasting: one BiGRU inference $O(W \cdot F \cdot H_u)$ per substation-day, multi-horizon adds factor K to the final dense layer only; training offline. Network LP: $|\mathcal{S}|K$ blocks ×~6 vars + $|\mathcal{L}|K$ trade vars ≈ 3·15·6 + 6·15 = **360 variables**, a few hundred constraints — milliseconds with HiGHS (`scipy.optimize.linprog`), solved once daily; **warm-starting** from the previous day's solution accelerates the rolling horizon. MOLP: closed-form bang-bang, $O(H)$ threshold evals + $O(T \cdot H)$ accounting. Pool: $O(|\mathcal{E}|)$ scoring + $O(|\mathcal{E}| \log N)$ TopK. KG retrieval: $O(|V_q| + |E_q|)$. LLM: external, bounded by the fixed prompt budget.

Caveats to state, not hide. (i) 15-day skill decays — publish the $R^2(k)$ curve; distant leads are advisory, only $k = 1$ is executed. (ii) "Trading" is scheduled inter-tie transfer with an exchange premium — no AC power-flow / OPF claims. (iii) The model assumes **cooperative SGOs under a common planning authority** — realistic only under a single regulator or an explicit exchange agreement; stated as a counterfactual. (iv) Design parameters ($\bar{P}_i$, battery size, $m, v, \delta_{\max}, K$) each get a **1-D sensitivity sweep** (shedding hours, surcharge months, SGO profit vs. parameter), and VoLL is reported for $v \in \{30, 60, 120\}$. (v) No transmission constraints beyond tie-line capacity and the §3.6 category rationing.

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12. End-to-end worked thread

A. Network coordination (executed block, $k = 1$):

Figure N1 — transfer on the peak day:

```
Kalyanpur (DESCO)                Dhanmondi (DPDC)
robust p̃ = 104.2 MW              robust p̃ = 172.4 MW
firm  P̄ = 140 MW                 firm  P̄ = 160 MW
surplus +35.8 MW   ──12.8 MW──▶  deficit -12.4 MW
                   (3% line loss)
books +153k BDT export           surcharge Δr ≈ 0.31 BDT/kWh (capped)
                                 vs shedding 4.5M BDT (44× surcharge)
```

Layer	Equation(s)	Value produced
Forecast	13–16 (BiGRU, K=15)	Dhanmondi $\tilde{p}_5$=172.4 MW (day-5 deficit flagged)
Deficit	19	Def = 12.4 MW > 0
Network LP	20–24	import $T_{K\to D}$=12.78 MW; battery held; LS=0
Premium	36–38	$\varphi^* = \varphi^{cap}$ → Δr≈0.31 BDT/kWh
SGO profit	39	Kalyanpur +153k BDT export
GCS	41–43 (η=emergency)	0.81 > 0.55 → enters pool, top-ranked

B. Household DR + billing (DHH0016, 2024-06-30 event, June 2026 bill):

Layer	Equation(s)	Value produced
Stress	19: $\hat{p} \geq \tau$	$s_d = 1$ → DR window
MOLP	30–31 (middle, α=0.60): shift ✓ curtail ✓	E^{sh}=4.864, E^{cu}=0.705 kWh
Rebate	ρE^{cu}	2.11 BDT
Bill	34–35 (June 2026)	4,647.5 → 3,924.4 BDT (S=723)
QoE	35 (May)	87.3
KG	§8	evidence graph, network + household nodes
LLM	50 grounding	hallucination-free explanation

Every number in a final customer explanation traces through this chain to either a real BPDB record or a stated equation — the formal content of "explainability over optimisation".

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13. Implementation map

1. `05b_substation_forecast.py` → multi-horizon head (Eq. 13), output dim K=15; add per-horizon metrics + σ_k (`05c_horizon_skill.csv`) and robust peaks (Eq. 16); extend

to Lalbag.

2. **New `06n_network_optimizer.py`** — receding-horizon LP (Eqs. 20–25, 38) with warm-start; outputs `06n_dispatch_log.csv`, `06n_transfer_events.csv`, `06n_surcharge_ledger.csv`.
3. **`06_molp_optimizer.py`** — unchanged logic; accept monthly surcharge $\Delta r_{i,m}$ (Eq. 26) as optional tariff input.
4. **`07_priority_pool.py`** — add NI feature (Eq. 40c) + regime-indexed weights & γ (Eqs. 41–43) as a config table; admit network events.
5. **`08_knowledge_graph.py`** — new Battery / TransferEvent / DeficitForecast / SurchargeRecord node & edge CSVs (§8).
6. **Docs/slides** — pipeline caption: "Deep Forecasting (BiGRU, 15-day) → Network Coordination LP → Household MOLP → Adaptive Priority Pool → Twin Adapter/KG → LLM"; relabel legacy "LSTM Forecasting".

End of consolidated model. Supersedes `mathematical_model.pdf` (v1), `mathematical_model_v2.md`, and `mathematical_model_v3_network_extension.md`.